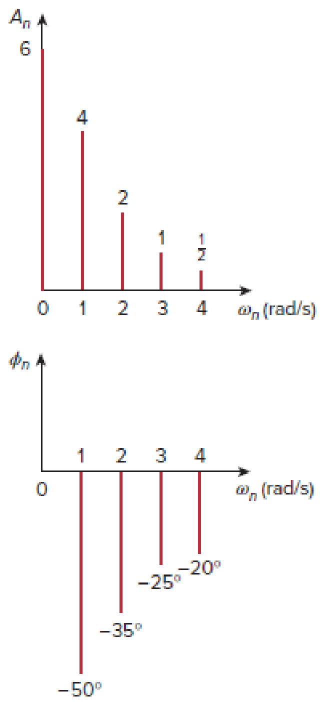
Problem

The spectra of the Fourier series of a function are shown in Fig. 17.84. (a) Obtain the trigonometric Fourier series, (b) Calculate the rms value of the function.

Figure 17.84:



Step-by-step solution

Step 1 of 2

a) From the wave form, $\omega_0 = 1$

The trigonometric fourier series is given by,

$$f(t) = a_0 + \Sigma A_n \cos(n\omega_0 t - \phi_n)$$

$$\therefore\ f(t) = 6 + 4\cos(t + 50°) + 2\cos(2t + 35°) + \cos(3t + 25°) + 0.5\cos(4t + 20°)$$

We have $\cos(A + B) = \cos A \cos B - \sin A \sin B$

$$f(t) = 6 + 4[\cos t \cos 50° - \sin t \sin 50°] + 2[\cos 2t\ \cos 35° - \sin 2t \sin 35°]$$

$$+[\cos 3t \cos 25° - \sin 3t \sin 25°] + 0.5[\cos 4t \cos 20° - \sin 4t \sin 20°]$$

$$\boxed{\begin{aligned} f(t) = 6 &+ 2.571\cos t - 3.064\sin t + 1.638\cos 2t - 1.147\sin 2t \\ &+ 0.9063\cos 3t - 0.423\sin 3t + 0.4698\cos 4t - 0.171\sin 4t \end{aligned}}$$

Step 2 of 2

b) The rms value of the function is given by

$$f_{rms} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} A_n^2}$$

$$f_{rms} = \sqrt{6^2 + \frac{1}{2}\left(4^2 + 2^2 + 1^2 + \left(\frac{1}{2}\right)^2\right)}$$

$$f_{rms} = \sqrt{36 + \frac{1}{2}(16 + 4 + 1 + 0.25)}$$

$$f_{rms} = \sqrt{36 + \frac{21.25}{2}}$$

$$f_{rms} = \sqrt{36 + 10.625}$$

$$f_{rms} = \sqrt{46.625}$$

$$\boxed{f_{rms} = 6.828}$$